

Complete splitting of planar Robin eigenvalues by a single shrinking hole

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Abstract

We study planar Robin eigenvalues when a scaled interior hole shrinks to a prescribed point, with the same fixed nonzero real Robin parameter on the outer and inner boundaries. We establish a complete multiscale description of every branch of a multiple limiting eigenvalue for arbitrary Lipschitz holes with connected exterior. The leading coefficients are Schur complements of an exterior harmonic energy matrix on the vanishing-order filtration. A nonnodal direction, when present, moves at first order with the sign of the Robin parameter; all nodal directions move down at even powers determined by their orders. We then construct an area-preserving analytic family of strictly convex holes near a disk. Outside a countable exceptional set of shape parameters, a single selected shape separates every fixed eigenvalue cluster, including at common nodal points. The smallness threshold for the hole size depends on the spectral window. Quantitative correctors, a two-term nodal formula and a nonnodal logarithmic correction accompany the splitting result.

1. Introduction and attribution

Small holes constitute a singular domain perturbation. A complete splitting theorem needs estimates that remain effective for eigenfunctions vanishing to different orders at the concentration point; one leading unweighted matrix estimate can miss entire branches. In the plane, the exterior Neumann corrector also requires a normalization modulo constants because a homogeneous Sobolev estimate does not control them.

The main result is stated in Section 2. Its proof uses the arbitrary-hole planar asymptotics in Sections 3-15 and an explicit variation of the exterior energy coefficients in Sections 16-19. The hole center remains fixed. The selected hole shape is independent of the spectral window, whereas its size threshold may depend on that window. We do not claim a uniform threshold for the entire infinite spectrum.

Felli-Roychowdhury-Siclari [FRS26] establish Robin spectral stability for either sign of the parameter and sharp simple-eigenvalue asymptotics, with a planar restriction to round holes for their nodal coefficient. Their final Remark 2.10 leaves multiple Robin eigenvalues and splitting for future work. Their perturbative framework, uniform extension and trace estimates are prior work. We include the needed compactness arguments and derive quantitative cutoff residuals to make the planar remainder explicit.

Felli-Liverani-Ognibene [FLO26] obtain generic reduction of multiplicity under removal of a spherical Neumann hole as its center varies, and complete splitting for double eigenvalues. Their arbitrary-hole leading matrix is proved in dimensions at least three; their planar theorem treats disks. Their earlier work [FLO25] supplies the related simple-eigenvalue analysis.

Abatangelo-Léna-Musolino [ALM26] correct the multiple-eigenvalue coefficients in the Dirichlet-hole theory of [ALM22]. Their corrected vanishing-order and matrix analysis is essential prior work: the projection method, filtration, and Schur-complement interaction are not claimed new here. The present calculations establish the Robin PDE coefficients and handle the possible coexistence of positive and negative shifts.

Shape differentiation of generalized polarization tensors and its Fourier interpretation are also prior work [AKLZ12]. Full splitting for other singular perturbations occurs in Gadyl'shin's boundary-patch problem [Gad93]. The distinction pursued here is the fixed-center, single-hole Robin conclusion with arbitrary finite multiplicity and explicit behavior at all nodal orders. Earlier expansion and repeated-eigenvalue frameworks [AKLZ10, LdC12, Dab17, Dab22] remain relevant. The inspected report [Dab17] gives leading matrix laws for grounded and conductivity inclusions; its stated applications do not give the fixed-center all-order Robin conclusion. The source-access limits of this comparison are disclosed in Section 20.

2. Main splitting theorem

Fix a bounded connected Lipschitz planar domain Ω , an interior point x_0 , and a nonzero real Robin parameter α . Translate x_0 to zero. Put

$$f(\theta) = \sum_{n=1}^{\infty} e^{-n^2} \cos(2n\theta), \quad \tilde{\Sigma}_t = \{(r, \theta) : 0 \leq r < 1 + tf(\theta)\}, \quad \Sigma_t = \left(\frac{\pi}{|\tilde{\Sigma}_t|} \right)^{1/2} \tilde{\Sigma}_t. \quad (\text{H1})$$

For all $|t| < t_*$, with an absolute sufficiently small $t_* > 0$, these are strictly convex real-analytic, centrally symmetric holes of area π . They contain zero and have connected exteriors. They converge to the unit disk in every fixed C^q norm as $t \rightarrow 0$. Indeed, the Fourier series and all its derivatives converge on each complex strip, f has zero average, and strict positivity of curvature persists in a small C^2 neighborhood of the circle. The area normalization factor is $1 + O(t^2)$.

Theorem 2.1 (complete fixed-center splitting). There is a countable set $\mathcal{N} \subset (-t_*, t_*)$, depending on Ω, x_0, α , such that the following holds. For every $t \notin \mathcal{N}$ and every eigenvalue λ of the unperforated Robin problem, there is $\varepsilon_\lambda(t) > 0$ such that every branch converging to λ is simple for

$$0 < \varepsilon < \varepsilon_\lambda(t)$$

in $\Omega \setminus \varepsilon \overline{\Sigma}_t$, with the same Robin parameter on both boundaries. In particular, for each finite M , there is $\varepsilon_M(t) > 0$ such that the first M perturbed eigenvalues are simple whenever $0 < \varepsilon < \varepsilon_M(t)$.

The center is fixed and may be a common nodal point of an eigenspace. The shape is fixed before $\varepsilon \rightarrow 0$, and can be chosen arbitrarily close to a disk. The threshold can depend on the spectral window. **No common positive threshold for the entire infinite spectrum is asserted.** The theorem is not uniform in α , in the center, or in the outer domain. For a single eigenvalue cluster, every sufficiently small nonzero t works, rather than merely all t outside a countable set.

The finite-window assertion also holds with any prescribed area in place of π , by a fixed scaling. The unperforated simple eigenvalues require no splitting argument: convergence and a spectral gap already give an isolated one-dimensional perturbed cluster.

3. Geometry and the simple nodal branch

Let $\Omega \subset \mathbb{R}^2$ be a bounded connected Lipschitz domain containing zero. Let Σ be a nonempty bounded Lipschitz open set with $\overline{\Sigma} \subset B_R$, and suppose $B_R \setminus \overline{\Sigma}$ is connected. We do not require that zero lie in Σ . Put

$$D = \mathbb{R}^2 \setminus \overline{\Sigma}, \quad \Omega_\varepsilon = \Omega \setminus \varepsilon \overline{\Sigma}.$$

Fix $\alpha \in \mathbb{R}$. Impose $\partial_\nu u + \alpha u = 0$ on every boundary component. The normal on a hole points into the hole. Let λ be a simple eigenvalue in Ω , with real eigenfunction u normalized by $\|u\|_{L^2(\Omega)} = 1$. Assume $u(0) = 0$, and write its interior Taylor expansion as

$$u(x) = P_k(x) + O(|x|^{k+1}), \quad \nabla u(x) = \nabla P_k(x) + O(|x|^k), \quad k \geq 1.$$

Here P_k is a nonzero homogeneous harmonic polynomial. Let W_k be the exterior harmonic function determined by

$$\partial_\nu W_k = \partial_\nu P_k \quad \text{on } \partial\Sigma, \quad W_k(y) = O(|y|^{-1}) \quad (|y| \rightarrow \infty). \quad (\text{S1})$$

Define

$$E_k = \int_D |\nabla W_k|^2, \quad C_k = E_k + \int_\Sigma |\nabla P_k|^2. \quad (\text{S2})$$

Theorem 3.1 (simple nodal branch). Let λ_ε be the eigenvalue converging to λ . Then

$$\lambda_\varepsilon = \lambda - C_k \varepsilon^{2k} + O(\varepsilon^{2k+1}). \quad (\text{S3})$$

In particular $C_k > 0$ and the eigenvalue decreases for sufficiently small holes, for either sign of α . If u_ε is normalized with positive overlap with u , and $\chi \in C_c^\infty(\Omega)$ equals one near zero, then for every $\eta \in (0, 1/2)$,

$$\left\| u_\varepsilon - u + \varepsilon^k \chi W_k(\cdot/\varepsilon) \right\|_{H^1(\Omega_\varepsilon)} = O_\eta(\varepsilon^{k+1-\eta}). \quad (\text{S4})$$

Consequently, for each fixed S with $\overline{\Sigma} \subset B_S$,

$$\varepsilon^{-k} u_\varepsilon(\varepsilon \cdot) \longrightarrow P_k - W_k \quad \text{in } H^1(B_S \setminus \overline{\Sigma}), \quad (\text{S5})$$

and

$$\varepsilon^{-2k} \|u_\varepsilon - u\|_{H^1(\Omega_\varepsilon)}^2 \longrightarrow E_k. \quad (\text{S6})$$

Constants can depend on the fixed geometry, eigenvalue, α , cutoff and η . No uniformity over families of domains or eigenvalues is claimed.

4. Uniform estimates and spectral convergence

The facts in this section are established in [FRS26]. We include their elementary scaling and compactness proofs to specify exactly which geometry and parameter assumptions are needed. The fixed-domain trace, extension, Sobolev and compact embedding theorems for bounded Lipschitz domains are used in their usual forms.

Let $A = B_R \setminus \bar{\Sigma}$ be connected, and fix a bounded extension from $H^1(A)$ to $H^1(B_R)$ that agrees with the original function on A . For $v \in H^1(\Omega_\varepsilon)$ apply this extension after rescaling to $v(\varepsilon \cdot)$ and subtracting its mean on A . Restore the mean after extension. Poincaré's inequality on A gives an extension $\mathcal{E}_\varepsilon v \in H^1(\Omega)$, equal to v off the hole, such that

$$\int_{\varepsilon\Sigma} |\nabla \mathcal{E}_\varepsilon v|^2 \leq C \int_{\varepsilon A} |\nabla v|^2, \quad \int_{\varepsilon\Sigma} |\mathcal{E}_\varepsilon v|^2 \leq C \int_{\varepsilon A} |v|^2 + C\varepsilon^2 \int_{\varepsilon A} |\nabla v|^2. \quad (\text{F1})$$

The gluing across the hole boundary is valid because the traces agree. Hence the extension is uniformly bounded in H^1 . The connectedness hypothesis is used in the Poincaré estimate; no connectedness of Σ itself is needed.

The fixed annular trace inequality, rescaled, yields

$$\|v\|_{L^2(\partial(\varepsilon\Sigma))}^2 \leq C\varepsilon^{-1} \|v\|_{L^2(\varepsilon A)}^2 + C\varepsilon \|\nabla v\|_{L^2(\varepsilon A)}^2. \quad (\text{F2})$$

Given $0 < \eta < 1/2$, apply planar Sobolev embedding to $\mathcal{E}_\varepsilon v$ with exponent $p = 2/\eta$ and then Hölder's inequality on εA . This gives

$$\|v\|_{L^2(\varepsilon A)}^2 \leq C_\eta \varepsilon^{2-2\eta} \|v\|_{H^1(\Omega_\varepsilon)}^2, \quad \|v\|_{L^2(\partial(\varepsilon\Sigma))}^2 \leq C_\eta \varepsilon^{1-2\eta} \|v\|_{H^1(\Omega_\varepsilon)}^2. \quad (\text{F3})$$

A fixed outer-boundary collar misses the hole. The trace inequality on this collar, with an arbitrarily small coefficient of the gradient term, implies that for every $\delta > 0$

$$\|v\|_{L^2(\partial\Omega)}^2 \leq \delta \|\nabla v\|_{L^2(\Omega_\varepsilon)}^2 + C_\delta \|v\|_{L^2(\Omega_\varepsilon)}^2. \quad (\text{F4})$$

One can obtain this by multiplying by a fixed cutoff equal to one near $\partial\Omega$ and applying the compact trace inequality on Ω ; the cutoff vanishes near the hole. Combining (F3)-(F4) gives constants $c, a, b > 0$, independent of sufficiently small ε , for which

$$a \|v\|_{H^1(\Omega_\varepsilon)}^2 \leq q_\varepsilon(v, v) := \int_{\Omega_\varepsilon} (|\nabla v|^2 + cv^2) + \alpha \int_{\partial\Omega_\varepsilon} v^2 \leq b \|v\|_{H^1(\Omega_\varepsilon)}^2. \quad (\text{F5})$$

This holds for either sign of the fixed real α . Choose the constants so that the analogous form on Ω is also coercive. The associated operators are positive self-adjoint with compact resolvent.

Lemma 4.1 (spectral convergence). Each ordered Robin eigenvalue of Ω_ε converges, counting multiplicity, to the corresponding eigenvalue of Ω .

Proof. For the upper bound in the minmax principle, restrict the first j eigenfunctions of Ω to Ω_ε . They are analytic near zero, so the removed volume and inner-boundary terms tend to zero uniformly on their finite-dimensional span. Their mass matrix tends to the identity. Thus $\limsup \lambda_j(\Omega_\varepsilon) \leq \lambda_j(\Omega)$.

For the reverse inequality take an orthonormal collection of the first j perforated-domain eigenfunctions. Their shifted energies are uniformly bounded by the upper bound and (F5). Their extensions are uniformly bounded in $H^1(\Omega)$ by (F1); along a subsequence they converge weakly in H^1 and strongly in L^2 to functions $v_1, \dots, v_j$. The extension mass inside the shrinking hole tends to zero by the uniform L^4 bound and its vanishing area. Therefore the limits are orthonormal.

On a fixed outer collar the trace converges strongly in $L^2(\partial\Omega)$, by compactness of the trace map. The inner-boundary quadratic term tends to zero by (F3), even when $\alpha < 0$. For any fixed coefficient vector z , weak lower semicontinuity on $\Omega \setminus \overline{B}_\rho$, followed by $\rho \downarrow 0$, gives

$$q_0 \left(\sum_{l=1}^j z_l v_l, \sum_{l=1}^j z_l v_l \right) \leq \liminf q_\varepsilon \left(\sum_{l=1}^j z_l v_l^\varepsilon, \sum_{l=1}^j z_l v_l^\varepsilon \right). \quad (\text{F6})$$

Here the mass and outer-boundary terms converge; excluding a fixed disk before taking the limit avoids charging the artificial extension's gradient energy to the original form. Choose the subsequence to realize the lower limit of the j th eigenvalue. The right side of (F6) is at most $\liminf (\lambda_j(\Omega_\varepsilon) + c) |z|^2$. Minmax on the j -dimensional limiting span proves the lower bound. This argument also proves convergence along the full sequence. $\square$

For any isolated limiting eigenvalue, its full finite-dimensional cluster is consequently separated by a uniform gap from the complementary spectrum. This is the only spectral convergence input needed below. Interior analyticity follows from the constant-coefficient equation $-\Delta u = \lambda u$, regardless of the regularity of the outer boundary.

5. The normalized exterior problem

Consider the Hilbert space

$$X = \{v \in H^1(D \cap B_S) \text{ for every sufficiently large } S : \nabla v \in L^2(D)\} / \mathbb{R}, \quad \|[v]\|_X = \|\nabla v\|_{L^2(D)}.$$

Completeness follows by selecting representatives with zero average on $A = B_R \setminus \overline{\Sigma}$. Poincaré inequalities on each connected $B_S \setminus \overline{\Sigma}$, with this fixed average, turn a gradient-Cauchy sequence into an H^1 -Cauchy sequence on every bounded truncation.

If $g \in L^2(\partial\Sigma)$ has integral zero, the functional

$$[v] \mapsto \int_{\partial\Sigma} g v$$

is well defined and bounded on X : subtract the average on A , and apply the trace and Poincaré inequalities on A . Lax-Milgram gives a unique class $[W]$ satisfying

$$\int_D \nabla W \cdot \nabla v = \int_{\partial\Sigma} g v \quad ([v] \in X). \quad (\text{S7})$$

For clarity, compactly supported H^1 test functions are dense in this gradient norm. First truncate a representative at heights $\pm M$; the gradient error tends to zero as $M \rightarrow \infty$. For fixed M , multiply by a radial cutoff equal to one on B_L , zero outside B_{L^2} , and linear in $\log r$ between these radii. Its squared gradient integral is $2\pi/\log L$. The cutoff error tends to zero by boundedness of the truncated function and the integrability of its gradient. Local smoothing in Lipschitz charts gives the corresponding density for smooth tests up to the boundary.

The solution is harmonic outside a fixed disk. Its Fourier expansion there has the constant mode, a possible logarithmic mode, and the modes $r^{\pm j} \cos(j\theta)$ and $r^{\pm j} \sin(j\theta)$. Finite Dirichlet energy excludes the logarithmic and growing modes. The remaining expansion shows that a unique representative satisfies

$$W(y) = O(|y|^{-1}), \quad \nabla W(y) = O(|y|^{-2}). \quad (\text{S8})$$

One may obtain the estimates by expanding on a circle outside the hole and using convergence of the decaying Fourier series beyond twice that radius. This removes the additive constant, which cannot be controlled by a homogeneous Sobolev inequality in dimension two.

For $g = \partial_\nu P_k$, the compatibility condition follows from

$$\int_{\partial\Sigma} \partial_\nu P_k = - \int_\Sigma \Delta P_k = 0.$$

Thus (S1) exists and is unique. Its energy is $E_k = \int_{\partial\Sigma} (\partial_\nu P_k) W_k$. It is nonzero: otherwise W_k is constant, hence $g = 0$; testing g against the trace of P_k would give

$$0 = \int_{\partial\Sigma} (\partial_\nu P_k) P_k = - \int_\Sigma |\nabla P_k|^2,$$

impossible for a nonconstant polynomial on a nonempty open set. In particular $C_k > 0$.

6. Quantitative cutoff correctors

Choose $c > 0$ so that, for all small ε ,

$$q_\varepsilon(v, w) = \int_{\Omega_\varepsilon} (\nabla v \cdot \nabla w + cvw) + \alpha \int_{\partial\Omega_\varepsilon} vw$$

is uniformly equivalent to the H^1 inner product. This follows from the fixed outer-boundary trace inequality and the small-hole trace estimate in Section 4 (see also [FRS26]). Set $\mu = \lambda + c > 0$. For every fixed $\eta \in (0, 1/2)$, the latter estimate is

$$\|v\|_{L^2(\partial(\varepsilon\Sigma))} \leq C_\eta \varepsilon^{1/2-\eta} \|v\|_{H^1(\Omega_\varepsilon)}. \quad (\text{S9})$$

Define V_ε by

$$q_\varepsilon(V_\varepsilon, v) = L_\varepsilon(v), \quad L_\varepsilon(v) = \int_{\partial(\varepsilon\Sigma)} (\partial_\nu u + \alpha u)v. \quad (\text{S10})$$

The explicit approximate corrector is

$$A_\varepsilon(x) = \varepsilon^k \chi(x) W_k(x/\varepsilon). \quad (\text{S11})$$

The cutoff equals one on the entire hole for small ε . On its boundary, the discrepancy between the data in (S10) and $\partial_\nu A_\varepsilon + \alpha A_\varepsilon$ has L^2 norm $O(\varepsilon^{k+1/2})$. Indeed, the leading normal derivative is exactly $\varepsilon^{k-1} \partial_\nu P_k$, the Taylor remainder is $O(\varepsilon^k)$, $u = O(\varepsilon^k)$ there, and the trace of W_k on $\partial\Sigma$ belongs to L^2 .

From (S8), scaling and the compact support of χ ,

$$\|A_\varepsilon\|_{L^2(\Omega_\varepsilon)} \leq C \varepsilon^{k+1} \sqrt{1 + |\log \varepsilon|}, \quad \|\nabla A_\varepsilon\|_{L^2(\Omega_\varepsilon)} = O(\varepsilon^k). \quad (\text{S12})$$

On the support of $\nabla \chi$, both $\varepsilon^k W_k(x/\varepsilon)$ and its x -gradient are $O(\varepsilon^{k+1})$. Since W_k is harmonic,

$$\|(-\Delta + c)A_\varepsilon\|_{L^2(\Omega_\varepsilon)} \leq C \varepsilon^{k+1} \sqrt{1 + |\log \varepsilon|}.$$

Integration by parts, (S9), and the boundary discrepancy therefore give

$$|L_\varepsilon(v) - q_\varepsilon(A_\varepsilon, v)| \leq C_\eta \varepsilon^{k+1-\eta} \|v\|_{H^1(\Omega_\varepsilon)}.$$

The boundary integration is valid weakly for the exterior solution; no H^2 regularity at a Lipschitz corner is required. Uniform coercivity and (S10) imply

$$\|V_\varepsilon - A_\varepsilon\|_{H^1(\Omega_\varepsilon)} \leq C_\eta \varepsilon^{k+1-\eta}. \quad (\text{S13})$$

In particular

$$\|V_\varepsilon\|_{L^2} = O_\eta(\varepsilon^{k+1-\eta}), \quad T_\varepsilon := q_\varepsilon(V_\varepsilon, V_\varepsilon) = \varepsilon^{2k} E_k + O(\varepsilon^{2k+1}). \quad (\text{S14})$$

For the energy estimate, the discarded far-field gradient energy in (S11) is $O(\varepsilon^{2k+2})$; the mass term is $O(\varepsilon^{2k+2}(1 + |\log \varepsilon|))$; the inner-boundary term is $O(\varepsilon^{2k+1})$; and the outer-boundary term vanishes. For the cross term, write $R_\varepsilon = V_\varepsilon - A_\varepsilon$ and use the exact identity

$$q_\varepsilon(R_\varepsilon, A_\varepsilon) = L_\varepsilon(A_\varepsilon) - q_\varepsilon(A_\varepsilon, A_\varepsilon).$$

The boundary residual paired with A_ε is $O(\varepsilon^{2k+1})$ by their boundary L^2 norms. The bulk residual paired with A_ε is $O(\varepsilon^{2k+2}(1 + |\log \varepsilon|))$ by (S12). Finally $q_\varepsilon(R_\varepsilon, R_\varepsilon) = O_\eta(\varepsilon^{2k+2-2\eta}) = o(\varepsilon^{2k+1})$. This establishes the sharper error in (S14) without discarding cancellation in the cross term.

7. Simple-eigenvalue reduction and normalization

Write $f_\varepsilon = u - V_\varepsilon$, with u restricted to the perforated domain. The weak equation and (S10) give

$$q_\varepsilon(f_\varepsilon, v) = \mu \langle u, v \rangle, \quad (\mathcal{A}_\varepsilon - \mu) f_\varepsilon = \mu V_\varepsilon, \quad (\text{S15})$$

where $\mathcal{A}_\varepsilon$ is the positive operator associated with q_ε . The second equality holds in L^2 , since the first places f_ε in its operator domain.

Spectral convergence and simplicity give a uniform gap between μ and every eigenvalue of $\mathcal{A}_\varepsilon$ except $\mu_\varepsilon = \lambda_\varepsilon + c$. Let Π_ε be the orthogonal projection onto that eigenspace, and put $r_\varepsilon = (I - \Pi_\varepsilon) f_\varepsilon$. Expanding (S15) in an orthonormal eigenbasis gives

$$\|r_\varepsilon\|_{H^1} \leq C \|V_\varepsilon\|_{L^2}. \quad (\text{S16})$$

The bound follows because $\sup_{j \neq n} \mu_j^\varepsilon / (\mu_j^\varepsilon - \mu)^2$ is uniformly finite, including the large eigenvalues.

The useful exact identity is

$$\mu \langle u, V_\varepsilon \rangle = L_\varepsilon(u) - T_\varepsilon, \quad (\text{S17})$$

obtained by interchanging the two arguments in q_ε and using (S10). Further,

$$L_\varepsilon(u) = - \int_{\varepsilon\Sigma} (|\nabla u|^2 - \lambda u^2) + \alpha \int_{\partial(\varepsilon\Sigma)} u^2. \quad (\text{S18})$$

Hence $L_\varepsilon(u) = O(\varepsilon^{2k})$ and $\langle u, V_\varepsilon \rangle = O(\varepsilon^{2k})$. Consequently

$$\|f_\varepsilon\|_{L^2}^2 = 1 - \int_{\varepsilon\Sigma} u^2 - 2 \langle u, V_\varepsilon \rangle + \|V_\varepsilon\|_{L^2}^2 = 1 + O(\varepsilon^{2k}).$$

Equations (S14) and (S16) show $\|\Pi_\varepsilon f_\varepsilon\|_{L^2} = 1 + O(\varepsilon^{2k})$. Normalize this projection with its positive overlap with u . Its H^1 norm is uniformly bounded, so

$$\|u_\varepsilon - f_\varepsilon\|_{H^1} = O_\eta(\varepsilon^{k+1-\eta}) + O(\varepsilon^{2k}) = O_\eta(\varepsilon^{k+1-\eta}). \quad (\text{S19})$$

This proves (S4) by (S13). It also avoids an unjustified passage from a global little-oh estimate to local rescaled L^2 convergence.

For the eigenvalue, calculate using (S15) and (S17):

$$q_\varepsilon(f_\varepsilon, f_\varepsilon) - \mu \|f_\varepsilon\|_{L^2}^2 = L_\varepsilon(u) - T_\varepsilon - \mu \|V_\varepsilon\|_{L^2}^2.$$

The contribution of r_ε to the left side has absolute value $O(\|V_\varepsilon\|_{L^2}^2)$ by (S16). Thus

$$(\lambda_\varepsilon - \lambda) \|\Pi_\varepsilon f_\varepsilon\|_{L^2}^2 = L_\varepsilon(u) - T_\varepsilon + O(\|V_\varepsilon\|_{L^2}^2). \quad (\text{S20})$$

Taylor expansion in (S18) yields

$$L_\varepsilon(u) = -\varepsilon^{2k} \int_\Sigma |\nabla P_k|^2 + O(\varepsilon^{2k+1}).$$

Insert this and (S14) in (S20). The extra errors are $O_\eta(\varepsilon^{2k+2-2\eta})$ and $O(\varepsilon^{4k})$; both are smaller than the error in (S3). This proves (S3).

For (S5), uniform extension into the hole and planar Sobolev embedding imply, for fixed S ,

$$\|v(\varepsilon \cdot)\|_{L^2(B_S \setminus \bar{\Sigma})} \leq C_{\eta, S} \varepsilon^{-\eta} \|v\|_{H^1(\Omega_\varepsilon)}. \quad (\text{S21})$$

Indeed, apply Hölder with exponent $p = 2/\eta$ on a set of area $O(\varepsilon^2)$ after extension. Applying (S21) to the error in (S4) and dividing by ε^k gives $O(\varepsilon^{1-2\eta})$. The rescaled gradient error is $O(\varepsilon^{1-\eta})$. The Taylor expansion completes (S5). Finally, (S4), (S12), and the energy calculation following (S14) prove (S6).

8. The second nodal coefficient

Let P_{k+1} denote the homogeneous Taylor term of degree $k+1$ in u , allowing $P_{k+1} = 0$. It is harmonic, since the right side of $-\Delta u = \lambda u$ begins at degree k . Denote its decaying exterior corrector by W_{k+1} , with the same definition as (S1). Put

$$\mathcal{C}(P_k, P_{k+1}) = \int_\Sigma \nabla P_k \cdot \nabla P_{k+1} + \int_D \nabla W_k \cdot \nabla W_{k+1}.$$

The two-term refinement is

$$\begin{aligned} \lambda_\varepsilon - \lambda = & -C_k \varepsilon^{2k} \\ & + \varepsilon^{2k+1} \left[\alpha \int_{\partial \Sigma} (P_k - W_k)^2 - 2\mathcal{C}(P_k, P_{k+1}) \right] + o(\varepsilon^{2k+1}). \end{aligned} \quad (\text{S22})$$

Here is the remainder calculation. With A_ε and R_ε as above, (S13) gives $q_\varepsilon(R_\varepsilon, R_\varepsilon) = o(\varepsilon^{2k+1})$. Consequently the exact identity

$$T_\varepsilon = 2L_\varepsilon(A_\varepsilon) - q_\varepsilon(A_\varepsilon, A_\varepsilon) + q_\varepsilon(R_\varepsilon, R_\varepsilon)$$

can be expanded to the next order using only the known corrector W_k :

$$\begin{aligned} L_\varepsilon(A_\varepsilon) &= \varepsilon^{2k} E_k + \varepsilon^{2k+1} \int_{\partial\Sigma} (\partial_\nu P_{k+1} + \alpha P_k) W_k + O(\varepsilon^{2k+2}), \\ q_\varepsilon(A_\varepsilon, A_\varepsilon) &= \varepsilon^{2k} E_k + \alpha \varepsilon^{2k+1} \int_{\partial\Sigma} W_k^2 + O(\varepsilon^{2k+2}(1 + |\log \varepsilon|)). \end{aligned}$$

Meanwhile (S18) gives

$$\begin{aligned} L_\varepsilon(u) &= -\varepsilon^{2k} \int_{\Sigma} |\nabla P_k|^2 \\ &\quad + \varepsilon^{2k+1} \left[-2 \int_{\Sigma} \nabla P_k \cdot \nabla P_{k+1} + \alpha \int_{\partial\Sigma} P_k^2 \right] + O(\varepsilon^{2k+2}). \end{aligned}$$

The L^2 spectral error in (S20) is $O_\eta(\varepsilon^{2k+2-2\eta}) = o(\varepsilon^{2k+1})$; the normalization error is $O(\varepsilon^{4k}) = o(\varepsilon^{2k+1})$. Finally, (S7) implies

$$\int_{\partial\Sigma} (\partial_\nu P_{k+1}) W_k = \int_D \nabla W_{k+1} \cdot \nabla W_k.$$

Substituting these identities into (S20) gives (S22).

If $\Sigma = -\Sigma$, uniqueness and the far-field normalization imply $W_j(-y) = (-1)^j W_j(y)$. The mixed term in (S22) then vanishes by parity. For a centrally symmetric hole the next coefficient is therefore exactly $\alpha \int_{\partial\Sigma} (P_k - W_k)^2$.

9. The nonnodal logarithmic coefficient

Now assume $u_0 := u(0) \neq 0$ and $\alpha \neq 0$. Let

$$p = \mathcal{H}^1(\partial\Sigma), \quad h(y) = \nabla u(0) \cdot \nu(y) + \alpha u_0, \quad m = \int_{\partial\Sigma} h = \alpha u_0 p.$$

Choose a point $y_* \in \Sigma$ and put $F(y) = -(m/(2\pi)) \log |y - y_*|$. The normal convention gives $\int_{\partial\Sigma} \partial_\nu F = m$. The zero-mean argument of Section 5 supplies a decaying harmonic W with boundary data $h - \partial_\nu F$. Thus $U = F + W$ solves $\partial_\nu U = h$ and

$$U(y) = -\frac{m}{2\pi} \log |y| + O(|y|^{-1}).$$

Take the cutoff approximation

$$B_\varepsilon(x) = \varepsilon \chi(x) \left[U(x/\varepsilon) - \frac{m}{2\pi} \log \varepsilon \right]. \quad (\text{S23})$$

The subtraction is essential: on a fixed physical annulus the bracket is $-(m/(2\pi)) \log |x| + O(\varepsilon/|x|)$, with no growing constant. It follows by direct integration that

$$\|B_\varepsilon\|_{L^2} = O(\varepsilon), \quad q_\varepsilon(B_\varepsilon, B_\varepsilon) = \frac{m^2}{2\pi} \varepsilon^2 \log(1/\varepsilon) + O(\varepsilon^2). \quad (\text{S24})$$

To check the remainder in the energy: the harmonic logarithm contributes $m^2/(2\pi)$ per unit logarithmic radial length; its cross integral with the decaying part has a convergent tail; the fixed cutoff region contributes $O(\varepsilon^2)$; and the inner Robin term is $O(\varepsilon^3 \log^2 \varepsilon) = o(\varepsilon^2)$.

The bulk residual of (S23) is $O(\varepsilon)$ in L^2 . Its boundary discrepancy in (S10) has L^2 norm $O(\varepsilon^{3/2}(1 + |\log \varepsilon|))$. Estimate (S9) makes its dual norm $o(\varepsilon)$, for any fixed $\eta < 1/2$. Coercivity gives

$$\|V_\varepsilon - B_\varepsilon\|_{H^1} = O(\varepsilon), \quad \|V_\varepsilon\|_{L^2} = O(\varepsilon).$$

For the energy cross term, write $S_\varepsilon = V_\varepsilon - B_\varepsilon$ and use $q_\varepsilon(S_\varepsilon, B_\varepsilon) = L_\varepsilon(B_\varepsilon) - q_\varepsilon(B_\varepsilon, B_\varepsilon)$. The bulk residual pairs with B_ε to give $O(\varepsilon^2)$; the boundary pairing is $O(\varepsilon^3(1 + |\log \varepsilon|)^2)$. Also $q_\varepsilon(S_\varepsilon, S_\varepsilon) = O(\varepsilon^2)$. Therefore

$$T_\varepsilon = \frac{m^2}{2\pi} \varepsilon^2 \log(1/\varepsilon) + O(\varepsilon^2). \quad (\text{S25})$$

Equations (S16)-(S20) remain valid. This time $L_\varepsilon(u) = \alpha p u_0^2 \varepsilon + O(\varepsilon^2)$, $\langle u, V_\varepsilon \rangle = O(\varepsilon)$, and $\|\Pi_\varepsilon f_\varepsilon\|_{L^2}^2 = 1 + O(\varepsilon)$. It follows that

$$\lambda_\varepsilon - \lambda = \alpha p u_0^2 \varepsilon - \frac{\alpha^2 p^2 u_0^2}{2\pi} \varepsilon^2 \log(1/\varepsilon) + O(\varepsilon^2). \quad (\text{S26})$$

The first term is established in [FRS26]. Earlier small-Robin-hole and logarithmic-expansion results must also be compared before a priority claim for the second term.

10. Disk and ellipse coefficients

For $\Sigma = B_r$ and $P_k(\rho, \theta) = \rho^k (A \cos k\theta + B \sin k\theta)$,

$$W_k(\rho, \theta) = -r^{2k} \rho^{-k} (A \cos k\theta + B \sin k\theta), \quad C_k = 2k\pi r^{2k} (A^2 + B^2). \quad (\text{S27})$$

The coefficients use the homogeneous Taylor polynomial itself, including its factorial normalization.

For an ellipse with semiaxes $a, b > 0$, use exterior conformal coordinates

$$z = \frac{a+b}{2} \left(w + \frac{a-b}{a+b} w^{-1} \right), \quad |w| > 1.$$

For $P_1 = x$, the exterior solution is $W_1 = -b\rho^{-1} \cos \theta$ in $w = \rho e^{i\theta}$ coordinates. For $P_1 = y$ it is $W_1 = -a\rho^{-1} \sin \theta$. The common conformal metric factor cancels from the normal-data equality; Dirichlet energy is conformally invariant. Thus for $P_1 = \xi x + \zeta y$,

$$C_1 = \pi b(a+b)\xi^2 + \pi a(a+b)\zeta^2. \quad (\text{S28})$$

This is a useful check that arbitrary-hole coefficients need not depend only on area. It is not counted as an independent discovery or paper.

11. Multiple eigenvalues and their vanishing orders

Use the geometry and normal convention of Section 3 and the estimates of Section 4. Fix $\alpha \neq 0$. Let λ be an eigenvalue of multiplicity m on Ω , and let E be its real eigenspace with the $L^2(\Omega)$ inner product. Write $p = \mathcal{H}^1(\partial\Sigma)$ and

$$E^+ = \{u \in E : u(0) = 0\}.$$

If evaluation at zero is nonzero on E , choose the unit vector u_0 orthogonal to E^+ with $a = u_0(0) > 0$. Otherwise set $E^+ = E$ and omit u_0 .

For integers $k \geq 1$, let $F_k \subset E^+$ consist of functions whose derivatives of every order below k vanish at zero. Thus $F_1 = E^+$. Set $H_k = F_k \ominus F_{k+1}$, using the L^2 orthogonal complement inside F_k . Discard zero spaces and enumerate the remaining degrees as

$$1 \leq k_1 < \cdots < k_s, \quad d_r = \dim H_{k_r}.$$

This list is finite and $E^+ = \bigoplus_{r=1}^s H_{k_r}$. Indeed, the descending finite-dimensional filtration eventually stabilizes; its intersection is zero by interior analyticity and unique continuation. Any nonzero vector of H_{k_r} has exactly order k_r , and the map taking it to its homogeneous leading Taylor polynomial is injective. In two dimensions this gives $d_r \leq 2$.

Choose an orthonormal basis u_i adapted to these spaces. Denote its orders by κ_i (in increasing groups) and its leading harmonic polynomials by P_i . For a nonconstant harmonic polynomial P , let $W(P)$ be the decaying exterior solution of

$$\Delta W = 0 \text{ in } D, \quad \partial_\nu W = \partial_\nu P \text{ on } \partial\Sigma.$$

Define the bilinear form and its matrix

$$\mathcal{C}(P, Q) = \int_\Sigma \nabla P \cdot \nabla Q + \int_D \nabla W(P) \cdot \nabla W(Q), \quad C_{ij} = \mathcal{C}(P_i, P_j). \quad (\text{M1})$$

The matrix C is positive definite. To see this, a zero quadratic form forces $\sum_i t_i P_i$ to be constant on an open set, hence constant everywhere. Its homogeneous terms have positive degrees, so it is zero. Within each degree, injectivity of the leading-polynomial map gives all $t_i = 0$.

Partition C by the groups H_{k_r} . Let $C_{<r, <r}$ denote its principal block containing all smaller orders. Define

$$S_1 = C_{11}, \quad S_r = C_{rr} - C_{r, <r} C_{<r, <r}^{-1} C_{<r, r} \quad (r > 1). \quad (\text{M2})$$

Every S_r is positive definite. Its eigenvalues $s_{r,1}, \dots, s_{r,d_r}$ are independent of the orthonormal bases chosen in each H_{k_r} .

Theorem 11.1 (all nodal branches). The perturbed cluster admits a labeling with one branch for each pair (r, j) , satisfying

$$\lambda_{r,j}(\varepsilon) = \lambda - s_{r,j} \varepsilon^{2k_r} + o(\varepsilon^{2k_r}). \quad (\text{M3})$$

If evaluation on E is identically zero these are all m branches. Otherwise they are $m - 1$ branches, and the remaining branch satisfies

$$\lambda_0(\varepsilon) = \lambda + \alpha p a^2 \varepsilon + o(\varepsilon). \quad (\text{M4})$$

Thus for sufficiently small holes, every nodal branch decreases. If evaluation is nonzero, positive α gives exactly one increasing branch and $m - 1$ decreasing branches; negative α gives m decreasing branches. These assertions concern shifts relative to the limiting λ , not monotonicity as a function of hole size. For $E^+ = 0$, only (M4) occurs. No claim of complete simplicity is made when a block S_r has repeated eigenvalues.

The all-nodal assertion also holds for $\alpha = 0$, using Neumann spectral convergence. The mixed Neumann case is excluded: its nonnodal branch has a different scale and can couple with degree-one data at leading order.

12. Weighted reduction for a full cluster

Fix a coercive shift c and let $\mu = \lambda + c > 0$. All scalar products below are over Ω_ε . For any $u \in E$ put

$$L_u(v) = \int_{\partial(\varepsilon\Sigma)} (\partial_\nu u + \alpha u)v, \quad q_\varepsilon(V_u, v) = L_u(v), \quad f_u = u - V_u. \quad (\text{M5})$$

Let Π_ε be the orthogonal projection onto the full m -dimensional cluster converging to λ , for the shifted operator. Define $g_u = \Pi_\varepsilon f_u$ and $r_u = f_u - g_u$. Spectral convergence gives a uniform gap to the complementary spectrum. As in Section 7,

$$(\mathcal{A}_\varepsilon - \mu)f_u = \mu V_u, \quad \|r_u\|_{H^1} \leq C\|V_u\|_{L^2}. \quad (\text{M6})$$

Since $f_u \rightarrow u$ and $r_u \rightarrow 0$ in L^2 uniformly on the unit sphere of the fixed finite-dimensional space E , $u \mapsto g_u$ is an isomorphism onto the cluster for small ε .

In the selected basis, the cluster shifts are exactly the generalized eigenvalues of the symmetric pair

$$A_{ij} = q_\varepsilon(g_{u_i}, g_{u_j}) - \mu \langle g_{u_i}, g_{u_j} \rangle, \quad G_{ij} = \langle g_{u_i}, g_{u_j} \rangle, \quad G \longrightarrow I. \quad (\text{M7})$$

Write $b_\varepsilon = q_\varepsilon - \mu \langle \cdot, \cdot \rangle$. The following exact identity, rather than an unweighted norm estimate, keeps the information at every vanishing order:

$$A_{ij} = L_{u_i}(u_j) - q_\varepsilon(V_{u_i}, V_{u_j}) - \mu \langle V_{u_i}, V_{u_j} \rangle - b_\varepsilon(r_{u_i}, r_{u_j}). \quad (\text{M8})$$

To verify it, symmetry and the eigenfunction equation give

$$\mu \langle u_j, V_{u_i} \rangle = L_{u_i}(u_j) - q_\varepsilon(V_{u_i}, V_{u_j}).$$

Expand $b_\varepsilon(f_{u_i}, f_{u_j})$ and subtract the complementary spectral contribution. The cross terms between the cluster and its complement vanish for both q_ε and the mass inner product. Also

$$L_{u_i}(u_j) = - \int_{\varepsilon\Sigma} (\nabla u_i \cdot \nabla u_j - \lambda u_i u_j) + \alpha \int_{\partial(\varepsilon\Sigma)} u_i u_j. \quad (\text{M9})$$

For a nodal basis vector define

$$Z_i = \varepsilon^{\kappa_i} \chi W(P_i)(\cdot/\varepsilon), \quad R_i = V_{u_i} - Z_i.$$

Section 6, whose corrector estimates did not use simplicity, gives for fixed $0 < \eta < 1/2$

$$\|Z_i\|_{H^1} = O(\varepsilon^{\kappa_i}), \quad \|R_i\|_{H^1} = O_\eta(\varepsilon^{\kappa_i+1-\eta}), \quad \|V_{u_i}\|_{L^2} = O_\eta(\varepsilon^{\kappa_i+1-\eta}). \quad (\text{M10})$$

Scaling the two leading terms in (M8), and using Cauchy-Schwarz on their remainders, yields for any two nodal indices

$$A_{ij} = -\varepsilon^{\kappa_i+\kappa_j} (C_{ij} + o(1)). \quad (\text{M11})$$

For example the energy cross remainder is $O_\eta(\varepsilon^{\kappa_i+\kappa_j+1-\eta})$; the mass and complementary terms in (M8) are $O_\eta(\varepsilon^{\kappa_i+\kappa_j+2-2\eta})$. The finitely many entrywise errors in (M11) are uniform. In particular, with $D_\varepsilon = \text{diag}(\varepsilon^{\kappa_i})$, the whole nodal block is

$$A_{++} = -D_\varepsilon(C + o(1))D_\varepsilon. \quad (\text{M12})$$

This stronger statement cannot be replaced by a single unweighted $o(\varepsilon^{2k_1})$ bound.

13. The nonnodal direction in a multiple cluster

Assume u_0 is present. The logarithmic corrector in Section 9 gives

$$\|V_{u_0}\|_{L^2} = O(\varepsilon), \quad \|V_{u_0}\|_{H^1} = O(\varepsilon\sqrt{1+|\log \varepsilon|}), \quad A_{00} = \alpha p a^2 \varepsilon + o(\varepsilon). \quad (\text{M13})$$

For a nodal index j , equation (M9) gives $L_{u_0}(u_j) = O(\varepsilon^{\kappa_j+1})$. For the mixed corrector energy, use its defining equation to obtain the sharper bound

$$q_\varepsilon(V_{u_0}, V_{u_j}) = L_{u_0}(Z_j) + q_\varepsilon(V_{u_0}, R_j) = O(\varepsilon^{\kappa_j+1}). \quad (\text{M14})$$

Indeed the first boundary integral has data of order one, trace $\varepsilon^{\kappa_j}W(P_j)$ and length of order ε . The second term is $O_\eta(\varepsilon^{\kappa_j+2-\eta}\sqrt{1+|\log \varepsilon|})$, which is smaller. The remaining two terms in (M8) are bounded by $O_\eta(\varepsilon^{\kappa_j+2-\eta})$. Hence

$$A_{0j} = O(\varepsilon^{\kappa_j+1}). \quad (\text{M15})$$

Put $d = A_{00}$ and $v = (A_{0j})$. For small ε , $d \neq 0$. The change of basis

$$T_0 = \begin{pmatrix} 1 & -d^{-1}v \\ 0 & I \end{pmatrix}$$

tends to the identity and gives the exact congruence

$$T_0^T A T_0 = \begin{pmatrix} d & 0 \\ 0 & A_{++} - d^{-1}v^T v \end{pmatrix}. \quad (\text{M16})$$

The correction in its nodal block has entries $O(\varepsilon^{\kappa_i+\kappa_j+1})$. Thus it does not change the limiting matrix C in (M12), for either sign of d . The transformed mass matrix still tends to I .

14. Schur complements and signed shifts

For completeness we prove the finite-dimensional step, in a block form of the method already supplied by [ALM26]. Let $C(\varepsilon) \rightarrow C > 0$. Gaussian block elimination in increasing order gives an upper block-triangular matrix $U(\varepsilon)$, with identity diagonal blocks, for which

$$U(\varepsilon)^T C(\varepsilon) U(\varepsilon) = \text{diag}(S_1(\varepsilon), \dots, S_s(\varepsilon)), \quad S_r(\varepsilon) \rightarrow S_r. \quad (\text{M17})$$

All matrices depend continuously on $C(\varepsilon)$ because its principal blocks are uniformly positive definite. Completing the square in the preceding blocks shows that the limiting S_r is exactly (M2).

Set $T_1 = D_\varepsilon^{-1}U(\varepsilon)D_\varepsilon$. Its block above the diagonal, in row r and column $t > r$, is $\varepsilon^{k_t-k_r}U_{rt}(\varepsilon)$, so $T_1 \rightarrow I$. It transforms the nodal matrix in (M12), or in (M16), exactly into

$$-\text{diag}(\varepsilon^{2k_1}S_1(\varepsilon), \dots, \varepsilon^{2k_s}S_s(\varepsilon)). \quad (\text{M18})$$

After both transformations the mass matrix is $I + o(1)$, whereas the stiffness matrix is the direct sum of (M18) and, if present, d . The ordinary eigenvalues of this block matrix have exactly the asymptotics (M3)-(M4).

Here is a justification that replacing I by $I + o(1)$ preserves those asymptotics even when the matrix is indefinite. For a positive definite mass matrix B with $(1 - \delta)I \leq B \leq (1 + \delta)I$, consider a negative eigenvalue $h_j < 0$ of a real symmetric matrix H . The number of generalized eigenvalues below $-t$ is the negative inertia of $H + tB$. Since

$$H + t(1 - \delta)I \leq H + tB \leq H + t(1 + \delta)I,$$

monotonicity of eigenvalues in the matrix order, or its minmax proof, bounds the corresponding negative generalized eigenvalue θ_j by

$$\frac{h_j}{1 - \delta} \leq \theta_j \leq \frac{h_j}{1 + \delta}. \quad (\text{M19})$$

Multiplicity is handled by allowing equality and then approaching thresholds from either side. Applying the same statement to $-H$ bounds the positive eigenvalues. Congruence preserves inertia, so there is no unaccounted change in their numbers. Letting $\delta \rightarrow 0$ proves multiplicative equivalence separately for every nonzero eigenvalue, however small its scale. This proves Theorem 11.1.

15. Common-order clusters at second order

Suppose every nonzero $u \in E$ has the same positive order k . Thus evaluation vanishes identically, $s = 1$, and $m \leq 2$. Write

$$u_i = P_i + Q_i + O(|x|^{k+2}),$$

where P_i and Q_i are the homogeneous Taylor polynomials of degree k and $k + 1$. Both are harmonic, and Q_i may be zero. Define

$$B_{ij} = \alpha \int_{\partial\Sigma} (P_i - W(P_i))(P_j - W(P_j)) - \mathcal{C}(P_i, Q_j) - \mathcal{C}(Q_i, P_j). \quad (\text{M20})$$

Theorem 15.1 (common-order second coefficient). For each eigenvalue c_* of C , let $\mathcal{E}_{c_*}$ be its Euclidean eigenspace, and let $b_1, \dots, b_q$ be the eigenvalues of the compression of B to $\mathcal{E}_{c_*}$. The corresponding branches satisfy

$$\lambda_j(\varepsilon) = \lambda - c_*\varepsilon^{2k} + b_j\varepsilon^{2k+1} + o(\varepsilon^{2k+1}). \quad (\text{M21})$$

To prove this, use the bilinear energy identity

$$q_\varepsilon(V_{u_i}, V_{u_j}) = L_{u_i}(Z_j) + L_{u_j}(Z_i) - q_\varepsilon(Z_i, Z_j) + q_\varepsilon(R_i, R_j). \quad (\text{M22})$$

Its last term is $O_\eta(\varepsilon^{2k+2-2\eta}) = o(\varepsilon^{2k+1})$; expanding the first three terms gives, exactly as in the two-term calculation of Section 8,

$$A = \varepsilon^{2k}(-C + \varepsilon B + o(\varepsilon)). \quad (\text{M23})$$

In more detail, the terms containing Q_i, Q_j are the two mixed interior energies from (M9) plus their exterior energies, using $\int_{\partial\Sigma}(\partial_\nu Q_i)W(P_j) = \int_D \nabla W(Q_i) \cdot \nabla W(P_j)$. The Robin terms combine as the boundary square in (M20). Bulk mass, cutoff and complementary projection terms are all smaller than ε^{2k+1} .

We also need more than $G \rightarrow I$ here. The exact identity preceding (M9) gives $\langle u_i, V_{u_j} \rangle = O(\varepsilon^{2k})$. Expanding $\langle f_{u_i}, f_{u_j} \rangle$ and subtracting $\langle r_{u_i}, r_{u_j} \rangle$ gives

$$G = I + O(\varepsilon^{2k}) = I + o(\varepsilon). \quad (\text{M24})$$

Conjugating (M23) by $G^{-1/2}$ therefore leaves its first two coefficients unchanged. Finally diagonalize C . In a fixed eigenspace of C , the first perturbation is the compression of B ; the coupling to the orthogonal complement contributes $O(\varepsilon^2)$ after a block elimination across the fixed spectral gaps of C . Equivalently, the Schur complement of $-C + \varepsilon B - zI$ at $z = -c_* + \varepsilon b$ is $\varepsilon(B|_{\mathcal{E}_{c_*}} - bI) + o(\varepsilon)$. Continuity of the eigenvalues of these finite symmetric matrices gives (M21).

For a centrally symmetric hole, the two mixed terms in (M20) vanish by parity. This does not automatically split a repeated c_* : the compressed boundary form must itself have distinct eigenvalues.

16. Analytic dependence on hole shape

For a harmonic polynomial P without a constant term, let $W_{t,P}$ solve the exterior problem of Section 5 on $D_t = \mathbb{R}^2 \setminus \overline{\Sigma}_t$. Thus $W_{t,P}$ decays at infinity and its normal derivative into the hole equals that of P . Define

$$\mathcal{C}_t(P, Q) = \int_{\Sigma_t} \nabla P \cdot \nabla Q + \int_{D_t} \nabla W_{t,P} \cdot \nabla W_{t,Q}. \quad (\text{H2})$$

For each fixed pair P, Q , this is a real-analytic function of t on a sufficiently small fixed interval $(-t_*, t_*)$.

Here are details that address the two-dimensional constant ambiguity. Choose a smooth diffeomorphism Φ_t of the plane, mapping D_0 to D_t , equal to the radial boundary displacement in a collar of the circle, and equal to the identity outside B_R . The area-normalized displacement is real analytic in t with coefficients smooth in space; a radial cutoff supplies such a map, uniformly invertible after decreasing t_* . Pull back the weak problem to

$$X = \{v \in H_{\text{loc}}^1(\overline{D}_0) : \nabla v \in L^2(D_0)\} / \mathbb{R}.$$

The notation means H^1 on every bounded exterior truncation, including its inner boundary. Its norm is the gradient norm. The pulled-back form has coefficient

$$A_t = |\det D\Phi_t| (D\Phi_t)^{-1} (D\Phi_t)^{-T}.$$

It is uniformly coercive and real analytic as a bounded operator on X . The boundary functional is

$$\ell_{t,P}(v) = \int_{\partial\Sigma_t} (\partial_{\nu_t} P)(v \circ \Phi_t^{-1}) ds.$$

It has integral-zero data, since P is harmonic. Subtracting a fixed annular average and using the trace bound makes it a bounded functional on X . In the pulled-back boundary parametrization its coefficients are analytic in t in every finite smooth norm, so it is analytic in X^* . The inverse of the coercive operator is locally analytic by its norm-convergent Neumann series. This proves analytic dependence of the solution class.

Because Φ_t is the identity outside B_R , every representative is harmonic there. Subtract its mean on one fixed exterior circle. Finite Dirichlet energy excludes growing modes and a logarithm; this subtraction selects exactly the decaying representative. The circle mean is a bounded functional on anchored representatives, so this normalization preserves analytic dependence on bounded truncations. The energy in (H2) can be written as the pulled-back bilinear form applied to the solution classes; the interior integral follows by change of variables. This proves the asserted analyticity of $\mathcal{C}_t$ without a planar homogeneous Sobolev inequality.

We will differentiate a boundary equation at $t = 0$. This is justified by pulling it back to the fixed smooth circle: the differentiated weak solution satisfies a smooth, uniformly elliptic problem in a fixed collar. Difference quotients converge in H^1 , and local Neumann regularity upgrades this convergence in each fixed finite Sobolev order after applying the differentiated equation. Equivalently one may differentiate the coefficients and boundary data in the pulled-back variational equation first. Outside B_R the Fourier coefficients are continuous linear functionals of the solution traces and therefore analytic as well.

17. Shape variation at the disk

It suffices to differentiate the unnormalized radial family: its normalization changes the shape only at order t^2 . Put

$$U_{t,P} = P - W_{t,P}.$$

This has zero normal derivative on the hole, with $U_{t,P} - P = O(|x|^{-1})$ at infinity. For

$$P(r, \theta) = r^k p(\theta), \quad p(\theta) = a \cos k\theta + b \sin k\theta,$$

the disk solution is

$$U_{0,P}(r, \theta) = (r^k + r^{-k})p(\theta). \tag{H3}$$

Let $\dot{U}_P$ be the Eulerian derivative at zero. It is harmonic for $r > 1$ and decays at infinity. Differentiate the zero normal derivative on the radial graph. Multiplying by the harmless normalizing factor, that equation reads

$$\partial_r U_{t,P}(1 + tf, \theta) - \frac{tf'}{(1 + tf)^2} \partial_\theta U_{t,P}(1 + tf, \theta) = 0.$$

Using $\partial_{rr} U_{0,P}(1, \theta) = 2k^2 p(\theta)$ and $\partial_\theta U_{0,P}(1, \theta) = 2p'(\theta)$ gives

$$\partial_r \dot{U}_P(1, \theta) = -2k^2 f p + 2f' p' = 2(fp')'. \tag{H4}$$

The right-hand side has mean zero, so it has a unique decaying exterior Neumann solution. Notice that the radial derivative here points out of the hole; the zero condition itself is independent of that sign choice.

We now relate a decaying multipole to (H2), to avoid assuming a shape-derivative formula. If Q is harmonic and $P = r^k p$, Green's identity on $B_R \setminus \bar{\Sigma}_t$ yields

$$\mathcal{C}_t(P, Q) = \int_{\partial B_R} (P \partial_r W_{t,Q} - W_{t,Q} \partial_r P) ds. \quad (\text{H5})$$

Indeed the corresponding inner-boundary integral is $-\int_{\Sigma_t} \nabla P \cdot \nabla Q - \int_{D_t} \nabla W_{t,P} \cdot \nabla W_{t,Q}$. The second equality uses the exterior variational identity; all functions have the required zero-mean data and finite energy. If the k th decaying Fourier mode of $W_{t,Q}$ is $r^{-k} w_k(\theta)$, (H5) becomes

$$\mathcal{C}_t(P, Q) = -2k \int_0^{2\pi} p(\theta) w_k(\theta) d\theta. \quad (\text{H6})$$

For P and Q of the same degree k , write $Q = r^k q(\theta)$. The k th Fourier mode of $\dot{U}_Q$ on the circle is $-k^{-1} \text{proj}_k[2(fq')']$, by (H4). Since $\dot{W}_Q = -\dot{U}_Q$, differentiation of (H6) and integration by parts give

$$\left. \frac{d}{dt} \mathcal{C}_t(P, Q) \right|_{t=0} = -4 \int_0^{2\pi} p(fq')' = 4 \int_0^{2\pi} f p' q'. \quad (\text{H7})$$

This is the familiar tangential-energy shape variation in this setting. It is derived here for the needed disk case; shape differentiation of generalized polarization tensors is established prior literature and is not claimed new.

Only the constant and $2k$ Fourier components of f contribute to (H7). For a zero-mean f with $2k$ component $A_k \cos 2k\theta + B_k \sin 2k\theta$, the derivative matrix in the polynomial basis $(r^k \cos k\theta, r^k \sin k\theta)$ is

$$2\pi k^2 \begin{pmatrix} -A_k & -B_k \\ -B_k & A_k \end{pmatrix}. \quad (\text{H8})$$

For (H1), $A_k = e^{-k^2} > 0$ and $B_k = 0$. Its two eigenvalues are opposite and nonzero, for every positive integer k .

18. Complete splitting of every Schur block

Fix an unperforated eigenspace E and use the orthogonal vanishing-order decomposition of Section 11. This basis is fixed in Ω and does not depend on t . Let $C(t)$ be the matrix with entries $\mathcal{C}_t(P_i, P_j)$, and let $S_r(t)$ be its Schur blocks, eliminating smaller positive orders first. Analyticity and positive definiteness of $C(t)$ imply that every $S_r(t)$ is positive definite and analytic on the whole small interval under consideration.

At $t = 0$ the matrix $C(0)$ has no cross terms between distinct degrees, by orthogonality of Fourier modes on the disk and on its exterior. Consequently

$$S_r(0) = C_{rr}(0), \quad S'_r(0) = C'_{rr}(0). \quad (\text{H9})$$

The second identity follows by differentiating $C_{rr} - C_{r,<r} C_{<r,<r}^{-1} C_{<r,r}$: both outer factors of the correction vanish at zero.

A block of dimension one is already simple. For a block of dimension two, let k be its order and write its two leading polynomials in the standard harmonic basis using an invertible real coefficient matrix M . At the disk,

$$C_{rr}(0) = 2k\pi M^T M. \quad (\text{H10})$$

If this matrix has distinct eigenvalues, they stay distinct for sufficiently small t . Otherwise it equals cI for some $c > 0$, so $M = \sqrt{c/(2k\pi)} O$ with O orthogonal. Equations (H8)-(H10) then give

$$S'_r(0) = cke^{-k^2} O^T \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} O. \quad (\text{H11})$$

The eigenvalue difference of $S_r(t)$ is therefore

$$2cke^{-k^2}|t| + o(|t|). \quad (\text{H12})$$

In particular, it is positive for every sufficiently small nonzero t . There are finitely many blocks in E , so all become simple for every sufficiently small nonzero t . The nonnodal direction, when present, is one-dimensional and moves at order ε , while all positive-order blocks move at distinct even powers of ε . Thus it cannot collide with them for sufficiently small ε . Within a block, distinct positive Schur coefficients give distinct leading eigenvalue shifts by Theorem 11.1. This proves the assertion for a fixed eigenspace and every sufficiently small nonzero t .

For a two-dimensional block define its discriminant

$$\Delta_r(t) = (\text{tr } S_r(t))^2 - 4 \det S_r(t). \quad (\text{H13})$$

It is analytic and not identically zero, by (H10) or (H12). Its zero set in $(-t_*, t_*)$ is discrete, hence countable. Taking the finite union over the blocks of an eigenspace and then the countable union over the discrete unperforated spectrum gives a countable exceptional set $\mathcal{N}$. For every t outside that set, every Schur block of every eigenspace has simple coefficients. Apply Theorem 11.1 separately to each cluster. For a finite spectral window, take the minimum of the finitely many thresholds and also ensure that different limiting clusters remain separated by their positive limiting gaps. This proves Theorem 2.1.

If a finite window ends inside a multiple limiting cluster, include that entire cluster when taking this finite minimum. The result then implies simplicity for the requested first M eigenvalues.

19. An exact conformal coefficient calculation

There is a useful independent coefficient check for every positive integer k . Consider the exterior conformal map

$$z = \Psi_t(w) = w + tw^{-(2k-1)}, \quad |w| > 1. \quad (\text{H14})$$

For sufficiently small real t , the boundary is a smooth simple perturbation of the circle and the map is univalent on its exterior. For example, the boundary is embedded and the derivative has no zeros for small t ; the argument principle, together with the simple pole at infinity, gives degree one on the unbounded complementary component. Its first normal displacement is $\cos(2k\theta)$, though the boundary parametrization also has a tangential displacement. Its enclosed area is $\pi(1 - (2k-1)t^2)$ by the boundary area integral.

For $P = \text{Re } z^k$ one has

$$\Psi_t(w)^k = w^k + ktw^{-k} + \sum_{j=2}^k \binom{k}{j} t^j w^{-(2j-1)k}. \quad (\text{H15})$$

In these conformal coordinates the zero-Neumann total field is $\text{Re}(w^k + w^{-k})$, since every growing mode is w^k and each boundary radial derivative must vanish. Therefore the pulled-back corrector is

$$W = \text{Re} \left((kt - 1)w^{-k} + \sum_{j=2}^k \binom{k}{j} t^j w^{-(2j-1)k} \right).$$

The exterior Dirichlet energy is the sum of πn times the squared decaying Fourier coefficients. Green's identity on the hole gives the interior energy as πk minus that same weighted sum for the negative coefficients in (H15), namely the sum with kt rather than $kt - 1$ at the first mode. Adding the two energies cancels every higher mode and gives the exact formula

$$\mathcal{C}_t(\text{Re } z^k, \text{Re } z^k) = 2\pi k(1 - kt). \quad (\text{H16})$$

For $P = \text{Im } z^k$, the negative Laurent powers have the opposite sine sign on the boundary. The same calculation gives

$$\mathcal{C}_t(\text{Im } z^k, \text{Im } z^k) = 2\pi k(1 + kt), \quad \mathcal{C}_t(\text{Re } z^k, \text{Im } z^k) = 0. \quad (\text{H17})$$

Scaling the hole to area π multiplies both coefficients by $(1 - (2k - 1)t^2)^{-k}$. Thus the two derivatives at zero are exactly $-2\pi k^2$ and $2\pi k^2$, agreeing with (H8). This is an exact harmonic calculation, not a numerical PDE test. This one-mode conformal family is used to check a fixed degree; the all-degree construction in (H1) remains the family used in the theorem.

20. Scope, source comparison and review status

The main assertion is one coherent result about a fixed shrinking-hole family. Its simple-eigenvalue formulas, multiple-eigenvalue reduction, shape derivative and splitting consequence are not counted as separate papers. The mixed nonnodal Neumann case is excluded from Theorem 11.1. Complete simplicity at one common positive hole size for the whole infinite spectrum is not proved. No parameter-uniform or geometry-uniform threshold is asserted.

The source comparison uses the final June 2026 Robin article [FRS26], the inspected arXiv versions of [FLO25, FLO26, ALM26], and the openly accessible older Robin-hole statements [Oza92, OR92]. The 2017 report [Dab17], including its variational principle and all application statements, was also inspected. It is cited separately from [Dab22], whose final text was unavailable. The full text of [AKLZ10, AKLZ12, LdC12, Dab22, Gad93] has not been completely compared in this investigation. Their abstracts or primary indexed excerpts establish relevant prior frameworks but do not suffice to exclude theorem-level overlap. No absence from a search is treated as proof of novelty.

The originating assistant has checked the internal derivations, including the signs in the reduced matrix, the weighted remainders, the logarithmic normalization, and the countable-exception argument. This is not independent mathematical review. Markdown formula checks establish syntax only. The accompanying source audit records precisely which sources were read and which retrievals failed. The contribution is presented as one unreviewed preprint. The mathematical argument is a hand proof; the finite diagnostics are supplementary and are not proof dependencies. No guarantee of exhaustive priority or journal acceptance is made.

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